

SUBODH KUMAR

PROFESSIONAL SUMMARY

AI/ML Engineer with 4.5+ years of backend engineering experience and hands-on expertise in building AI-powered applications, predictive analytics systems, and production-oriented ML workflows. Skilled in FastAPI, Python, RAG pipelines, vector databases, FAISS indexing, and machine learning model integration. Experienced in developing context-aware AI assistants, predictive maintenance platforms, conversational APIs, and scalable backend systems.

Strong understanding of ML preprocessing, feature engineering, REST APIs, model serving, semantic retrieval, and AI system architecture. Proven experience working on real-world fintech, AI, and analytics-driven systems with focus on scalability, reliability, and deployment-ready engineering.

TECHNICAL SKILLS

Programming & Backend

- Python
- PHP (Laravel)
- SQL
- FastAPI
- REST APIs
- JWT Authentication

AI / Machine Learning

- Scikit-learn
- Pandas
- NumPy
- Feature Engineering
- Classification & Regression
- Predictive Analytics
- Model Evaluation
- EDA
- Time-Series Forecasting

Artificial Intelligence / Machine Learning Engineer

FastAPI | RAG Pipelines | Predictive Analytics | Python Backend Systems

✈ Greater Noida U.P. India | ☎ +91-7007456072 | ✉ subodh.24fd@gmail.com
📄 <https://www.linkedin.com/in/subodh-kumar-76214a168/> | 🌐 <https://github.com/ksubodh9>

PROFESSIONAL EXPERIENCE

Vayam Technologies Ltd.

August 2023 – Present

Product Engineer / Software Engineer

MaddoxPay Fintech Platform

Backend Systems & Product Ownership

- Designed and maintained high-volume recharge and bill-payment systems handling 100K+ monthly transactions.
- Implemented automated payout systems (₹1M+ daily) using Axis Bank integrations including NEFT, RTGS, and IMPS.
- Developed secure fintech workflows with reconciliation systems and transaction monitoring.
- Optimized database schemas and backend workflows, improving response performance by nearly 2x.
- Built analytics dashboards and reporting systems for operational monitoring and risk analysis.

Applied AI / ML Integration

- Integrated predictive machine learning models into internal dashboards for anomaly detection and operational insights.
- Developed backend APIs for model serving and prediction workflows.
- Worked on deployment-ready AI integrations with reliability-focused fallback handling.
- Supported AI-enabled decision workflows through predictive analytics and backend automation.

Security & Infrastructure

- Implemented MFA, geofencing, secure sessions, biometric eKYC integrations, and fraud prevention workflows.
- Managed production deployments and backend services on Windows and CentOS environments.

Flyde Digital Solutions

July 2022 – August 2023

Fullstack Engineer

- Delivered backend-focused full-stack applications with scalable API integrations and optimized workflows.
- Integrated Razorpay, PhonePe, CC Avenue, and logistics APIs including Shiprocket and DTDC.
- Worked on AWS S3-based media storage systems and scalable backend architecture.
- Improved backend performance and database efficiency through query optimization and modular development.

Machine Learning Algorithms

- Linear Regression
- Logistic Regression
- Random Forest
- Decision Tree
- Gradient Boosting
- AdaBoost
- XGBoost
- ARIMA/SARIMA

Generative AI & NLP

- RAG (Retrieval-Augmented Generation)
- FAISS
- Vector Database
- Semantic Search
- Context-Aware Retrieval
- Conversational AI
- Embeddings
- Prompt Engineering

Visualization & Frontend

- Streamlit
- Matplotlib
- Seaborn
- Power BI (Basic)

Databases & Tools

- MySQL
- MSSQL
- MariaDB
- Git
- Docker
- Swagger UI

Saatirmind Software Services

May 2021 – July 2022

Laravel Developer

- Developed modular web applications using Laravel and MySQL.
- Worked on backend APIs, database optimization, and scalable application workflows.
- Improved system stability and deployment reliability through backend enhancements and debugging.

PROJECTS

Smart AI Assistant

Tech Stack: Laravel, Python, FAISS, Vector Database, RAG, NLP, LLM APIs

- Developed an AI-powered intelligent assistant with a custom Retrieval-Augmented Generation (RAG) pipeline for contextual question answering.
- Built the complete RAG workflow including:
 - Data collection and preprocessing
 - Embedding generation
 - FAISS vector indexing
 - Semantic retrieval pipeline
 - Context-aware response generation
- Implemented vector similarity search using FAISS for efficient knowledge retrieval.
- Integrated bilingual conversational support, contextual response handling, and file attachment processing.
- Worked on prompt orchestration, retrieval optimization, and scalable AI backend integration.

Career Guidance Engine API

Tech Stack: FastAPI, Python, RAG, FAISS, Vector Database, NLP, JWT, Voice APIs

- Built an AI-powered career guidance platform providing personalized recommendations and conversational career assistance.
- Developed a custom RAG pipeline for context-aware career guidance responses.
- Implemented semantic search, vector similarity retrieval, and intelligent recommendation workflows.
- Integrated FAISS indexing and embeddings for contextual information retrieval.
- Added voice assistant APIs for voice-enabled conversational interactions.
- Designed scalable FastAPI backend services with JWT authentication and modular architecture.

EDUCATION & CERTIFICATIONS

Executive Post Graduate Certification in Data Science & Artificial Intelligence

Intellipaat | 2025

Microsoft SQL Certification

Intellipaat | 2025

B.Tech – Information Technology

Dr. A.P.J. Abdul Kalam Technical University, Lucknow | 2020

Virtual Lab Development Program (Bootathon)

IIT Bombay | 2018

KEY STRENGTHS

- AI System Development
- RAG Pipelines & Semantic Search
- FastAPI Backend Engineering
- Predictive Analytics & ML Workflows
- Production-Oriented Backend Systems
- API Design & Model Serving
- Problem Solving & System Optimization
- Backend + AI Product Engineering

Industrial Equipment Monitoring & Predictive Maintenance System

Tech Stack: FastAPI, Streamlit, Python, Scikit-learn, Docker

- Engineered an AI-powered predictive maintenance platform for industrial equipment health monitoring.
- Built preprocessing and feature engineering pipelines for sensor-based industrial datasets.
- Implemented machine learning models for:
 - Fault Classification
 - Predictive Maintenance Forecasting
 - Equipment Failure Detection
- Experimented with Random Forest, XGBoost, Logistic Regression, and ensemble learning techniques.
- Developed FastAPI backend services and interactive Streamlit dashboards for real-time monitoring.
- Structured Docker-ready modular architecture for scalable deployment.

Customer Churn Prediction Platform

Tech Stack: Python, Scikit-learn, Pandas, NumPy, XGBoost

- Developed an end-to-end customer churn prediction system using supervised machine learning techniques.
- Trained and evaluated multiple models including Logistic Regression, Random Forest, Gradient Boosting, AdaBoost, Decision Tree, and XGBoost.
- Achieved highest accuracy of 80.3% using Logistic Regression after comparative model evaluation.
- Built reusable ML pipelines for preprocessing, evaluation, and prediction workflows.

House Price Prediction System

Tech Stack: FastAPI, Streamlit, Scikit-learn, Python

- Built a machine learning-powered house price prediction platform with FastAPI backend and Streamlit frontend.
- Implemented Linear Regression, Random Forest Regressor, and XGBoost Regressor models.
- Developed REST APIs for real-time inference and interactive prediction dashboards.
- Structured scalable backend architecture with reusable ML services.

Walmart Sales Forecasting System

Tech Stack: Python, Pandas, Scikit-learn, Matplotlib

- Developed a retail sales forecasting system using regression and time-series forecasting techniques.
- Implemented Linear Regression, Random Forest, XGBoost, and ARIMA/SARIMA-based forecasting workflows.
- Conducted trend, seasonality, and forecasting analysis using visualization and EDA techniques.

HR Analytics & Employee Attrition Prediction

Tech Stack: Python, Pandas, Scikit-learn, Matplotlib, Seaborn

- Developed an HR analytics system to analyze employee attrition patterns and workforce turnover.
- Implemented Logistic Regression and Decision Tree classification models.
- Generated actionable retention insights using EDA, feature engineering, and visualization-driven analysis.

Multimodal Biometric Authentication System

Tech Stack: Python, OpenCV, Computer Vision

- Developed a multimodal biometric authentication system using OpenCV and computer vision techniques.
- Implemented biometric verification workflows with image preprocessing and feature extraction.
- Explored AI-driven authentication and security use cases for real-world applications.